Hassan Mahmood

mahmood.h@northeastern.edu | linkedin.com/in/hassan-mahmood- | scholar.google.com/citations?user=LnjsphYAAAAJ

Education

Northeastern University Boston, MA
Ph.D. Computer Science (M.S. Degree Awarded in 2024) *Jan. 2027 (Expected)*
Advisor: Dr. Ehsan Elhamifar
Research Interests: Efficient Adversarial Attacks and Robustness, Generative Models, LLM Agent Evaluation

National University of Sciences and Technology Islamabad, Pakistan
Bachelor of Software Engineering *July 2018*
Advisor: Dr. Faisal Shafait
Research Interests: Computer Vision, Real-time Vehicle Detection

Publications

- **Hassan Mahmood**, Alina Oprea, Ehsan Elhamifar, "Clean-Label Backdoor Attacks for Diffusion Models", **(under review, NeurIPS 2026)**
- **Hassan Mahmood**, Ehsan Elhamifar, "Compositional Targeted Multi-Label Universal Perturbations", **CVPR 2025**. [PDF] [Code]
  Workshop Spotlight presentation — Uncertainty Quantification for Computer Vision Workshop (CVPR 2025).
- **Hassan Mahmood**, Ehsan Elhamifar, "Semantic-Aware Multi-Label Adversarial Attacks", **CVPR 2024**. [PDF] [Code]
- S. Qasim, **Hassan Mahmood**, Faisal Shafait, "Rethinking Table Recognition using Graph Neural Networks", **ICDAR 2019**. [PDF] [Code]

Work Experience

Ph.D. Intern May 2026 – Aug. 2026
Google Cloud AI *Seattle, WA*

Project: Multi-Agent LLM System for Fleet-Scale Root-Cause Analysis

- Designed and shipped specialized agents in a **hierarchical multi-agent** system automating root-cause analysis of security-patch failures across Google's fleet of **∼14M VMs**, a bottleneck in closing P0 vulnerabilities. Each subagent queries its own log sources and separates the originating fault from downstream symptoms by temporal event ordering, while an orchestrator reconciles them into one verdict for SRE remediation.
- Scaled from single-machine investigation to continuous fleet-wide coverage, reducing mean investigation time from **16h to ∼2h per 10K machines (8×)** and inconclusive cases from **20% to ∼8%** against the prior production system. **Deployed to SRE on-call** and recognized with a **Google AI Award, Gold (2026)**.

Graduate Research Assistant Aug. 2020 – Present
NOVA Lab, Northeastern University *Boston, MA*

Project: Action-State Inference in Multi-Task Egocentric Videos with Vision-Language Models (ongoing work)

- *Problem:* Inferring which actions of each task are complete at *any* point in interleaved **multi-task egocentric videos** requires the full action history. Dense sampling of that history far exceeds a VLM's context limit, while uniform sampling allocates by duration, oversampling long actions and leaving short ones with **zero frames**, so they are never recognized as completed.
- *Solution:* Built an action-coverage-aware pipeline that jointly selects frames and adaptively merges tokens under a **strict 3K visual token budget**, using no preset segment rules since the action count is unknown in advance. Recovers completion state for past actions of any queried task, raising **F1 from 0.68 to 0.73** over uniform sampling at equal budget.

Project: Clean-Label Backdoor Attacks on Diffusion Models (NeurIPS 2026, under review)

- *Problem:* Under a **data-only threat model**, naive small-perturbation poisons are low-density samples that the denoising process treats as noise and effectively ignores. Making them effective requires optimizing the poisons *through* the diffusion training dynamics, which is intractable to do exactly (**exhausts an H100's 94 GB** for even a single sample).
- *Solution:* Built **DDPF**, the first differentiable bilevel formulation for data-only diffusion poisoning, architecture-agnostic across attacker objectives, in which poisons are optimized so that standard diffusion training learns **corrupted score estimates** on them. Three surrogates (Tweedie single-step, marginal substitution, timestep-prior) make the bilevel objective tractable and **trainable in batch** where exact unrolling OOM'd on a single sample, raising attack success **∼20%** over heuristic baselines at a **5% poison rate**.

Project: Compositional Targeted Multi-Label Universal Perturbations (CVPR 2025)

- *Problem:* Targeted universal perturbations for multi-label models require solving for every target label combination, which is **exponential in label count** and makes full-label-space evaluation infeasible on datasets such as OpenImages.

- *Solution:* Built a compositional optimization framework that decomposes targeted perturbations under a **label-independence assumption**, reducing 2^n combinations to n per-label problems: **141h to 48h (2.9×)** wall-clock on 100K OpenImages samples over a 100-label subset (on V100 32 GB). Perturbations for unseen label combinations are generated in **constant time at inference**.

Project: Semantic-Aware Multi-Label Adversarial Attacks (CVPR 2024)

- *Problem:* Existing multi-label attacks fail to scale to large label spaces and corrupt predictions on correlated non-target labels.
- *Solution:* Identified **negative gradient correlation** between target and correlated labels as the root cause; built a **knowledge-graph** guided framework that generates semantic-aware perturbations and resolves the gradient correlation, achieving $\sim$ **20%** higher attack success rate and reduced runtime over prior attacks on large datasets (PASCAL-VOC, NUS-WIDE, OpenImages).

Project: Text-based Error Generation using LLM Prompt Tuning

- *Problem:* Generating realistic, valid error descriptions for procedural task videos is hard to control – standard **Llama-2** prompting yields invalid or low-quality errors with no reliable filter.
- *Solution:* Built an **adversarial discrete prompt-optimization** loop to generate realistic procedural-task errors with **Llama-2**, using a separate Llama-2 verifier to score and filter generations. Raised the fraction of valid error descriptions by ∼**35%** over baselines.

Red-Teaming Autonomous LLM Agents for Over-Trust Collapse

- Audited autonomous agents on Anthropic's **Petri** framework and showed that **Claude 3.7 Sonnet**, accountable for its judgment, cedes that judgment to a trusted tool as deadline pressure escalates. Built an **LLM feedback loop** that revises scenarios from judge summaries and retries them at higher pressure when the agent resists.

Graduate Research Assistant — Aug. 2019 – July 2020
CBL Lab, University of Houston — *Houston, TX*

Project: Explainable and Privacy-Aware Deep Learning

- Built cardiovascular disease prediction models on the **MESA** cohort, improving **F1 by 6%** over prior approaches, with **SHAP** and **LIME** attribution analyses to surface clinically interpretable risk factors.
- Designed an affective computing framework using temporal gait modeling over skeletal keypoints data, avoiding face and appearance capture for behavioral analysis; contributed directly to a **$100K** research grant award.

AI Researcher — July 2018 – Aug. 2019
Deep Learning Lab, National Center of Artificial Intelligence — *Islamabad, Pakistan*

Project: Table Structure Recognition using Graph Neural Networks

- *Problem:* Table structure recognition in document images fails to generalize across the varied, irregular layouts found in real-world document scans.
- *Solution:* Built a **Graph Neural Network** formulation modeling cell-to-cell structural relationships, improving accuracy **37%** over baselines. Open-sourced a synthetic tabular-document generator. ([Link]).

Engineering Skills

- **Programming Languages**: Python, C/C++, Java, SQL
- **ML Frameworks/Tools**: PyTorch, HuggingFace (Transformers, Diffusers), TensorFlow, OpenCV, NumPy
- **Training & Systems**: distributed data-parallel training (DDP), memory profiling and optimization, Git, Docker, Linux, GCP, AWS

Service

- Served as a reviewer for TPAMI journal, CVPR, AAAI, ICCV, and ICLR conferences.
- Supervised an undergraduate research intern for six months at University of Houston.

Awards

- Selected as one of the 190 students from a pool of 2500+ candidates across Pakistan for fully funded UGRAD semester exchange program sponsored by the U.S. Department of State (worth ≈ **$22,500**).
- Prime Minister National ICT R&D Merit Scholarship for fully funded undergraduate degree and monthly stipend for four years (worth ≈ **$12,000**).
- Bronze medal and merit scholarship for 3rd standing (in a total of ≈ **40,000** students) in computer science higher education exams in Pakistan - FBISE, Islamabad.